

PROJECT REPORT

"Week 1 Python Basics"

Project :

Create a Personal Introduction Program that asks for user's name, age, and hobby, then displays friendly welcome message with their information

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Project Overview

This report presents the work completed during **Week 1**, which focuses on introducing fundamental Python programming concepts and applying them through a simple web-based mini project. The objective of this task is to help beginners understand Python basics and gain initial exposure to backend development using the Flask framework.

Created a Flask-based web application that collects user details such as name, age, and hobby, and displays a friendly, personalized introduction message.

Objectives:

- To understand basic Python syntax and structure
- To practice arithmetic operations using Python
- To learn list creation, indexing, and iteration
- To introduce functions and modular coding
- To understand basic Flask routing and form handling
- To demonstrate client-server interaction

Tools and Technologies: Python, Flask, HTML, CSS, Jupyter Notebook

Setup and Installation Instructions

1. Install Flask using pip:

pip install flask

2. Maintain the following project structure:

Project Personal Intro Folder/

```
|—— app1.py
|—— static/
|   |—— shapes.jpeg
|—— templates/
|   |—— task1.html
|   |—— result.html
```

3. Run the application:
python app1.py
4. Open the browser and visit:
`http://127.0.0.1:5000/`

Code Structure Explanation

Backend (app1.py)

- Flask application initialization
- Route / to display the input form
- Route /result to process user input
- Uses POST method to collect form data
- Custom function generate_output() to generate dynamic output

Frontend

task1.html

User input form for name, age, and hobby.Sends data to backend using POST request.

result.html

Displays personalized greeting.Uses Jinja2 templating to render dynamic HTML safely.

Python Practice Notebook

The notebook includes:

- Hello World program
- Variable creation (name, age, city)
- Arithmetic operations using a function
- List creation, indexing, slicing, and looping

Functional Explanation

1. Python Basics:

- Variables store personal and user data
- Arithmetic operations demonstrate mathematical calculations
- Lists manage multiple values such as favorite foods

2. Personal Introduction Program

- User submits details via a web form
 - Flask server processes the data
 - Input is cleaned using .strip() and .split(',')
 - Hobbies are stored in a list and displayed concisely
 - A friendly, dynamic message is generated
-

Screenshots of the Application

Stage 1: Input Form

Get To Introduction Program

Stage 1: The Input Form

👋 **Welcome! Let's get acquainted.**

Just a few quick details to personalize your experience.

Name

Full Name

Age

28

📌 **What do you love to do for fun?**

Hiking, Painting, Coding

Say Hello! 🌟

Stage 2: Output Display

Your Personalized Introduction

🎉 **Fantastic to Meet You, Sanskruti Sahadev! 🎉**

Sanskruti Sahadev! 🎉

Here's a snapshot of who you are

- You are **Sanskruti Sahadev!** That's plenty of awesome to start!
- And we love that your favorite pastime is **Coding**. This seems like a wonderful way to spend your time.

Let's start the adventure! We're so happy to have you here. See just what we can do together.

Explore the Program

Go Back

Conclusion

The Week 1 project successfully fulfills all learning objectives. It demonstrates a clear understanding of Python fundamentals and introduces basic Flask web development concepts. The project serves as a strong foundation for advanced Python programming and full-stack development in future weeks.